

Ashwanth Fernando

Principal Software Engineer

Distributed Systems | Backend Engineering | AWS and GCP
Microservices | Product Focused and Customer First Mindset

New York / San Francisco

Australian Citizen

Email: ashwanthfernando@gmail.com

Phone: +1-415-763-9191

LinkedIn/ashwanthf

Summary

Hands-on principal-level engineer with **15+ years** building large-scale distributed systems and platform infrastructure at **Apple, Canva, eBay, Atlassian, and Walmart**. Scaled **Canva Search from 20M to 80M MAU** and built the team behind 5 live-collaboration products driving **10M+ MAU growth**; architected **99.999%-availability** global load balancers at Apple; drove Walmart's migration from a proprietary commerce platform to open source, **saving \$100M+ annually**. Author of **5 US patents**. Expert in Java, Go, and Python on AWS/GCP; currently shipping production agentic AI systems (LangGraph, RAG, LLM guardrails).

Work authorization: Australian citizen eligible for the E3 visa — no lottery, 2–4 week processing, minimal employer cost. Available onsite in New York or San Francisco, or remote from Sydney.

Technical Skills

Languages & Databases: Java (primary), Go, Python, Kotlin, Scala · MySQL, DynamoDB, Spanner, Redis, Elasticsearch, Solr

Cloud & Infrastructure: AWS, GCP, Kubernetes, Terraform, Docker, Spinnaker · Kafka, Spark, ZooKeeper, Temporal

Frameworks & AI: Spring, GraphQL, gRPC, Protobuf · LLMs, RAG, LangGraph, agentic workflows

Experience

Principal Engineer, Tentacle Software

Jan 2022 – Present

Boutique engineering consultancy for high-throughput, low-latency platforms — greenfield builds, architecture reviews, performance and reliability engineering. Engagements run 4–18 months.

Client: Macquarie Bank

24/02/2025 – Present

- Led a latency program across 5 teams (20+ microservices) on the Wealth Adviser Platform — fixed caching strategy, reactive processing, schema flaws, and JVM GC settings. **p95 down 14%, p99 down 9%, throughput up 3.5x**.
- Exposed mortgage serviceability services (income verification, expense assessment, policy lookup) as agent tools and tuned prompts across workflows, lifting eval scores **30%+**; patterns adopted by downstream squads.
- Built **Enforcer**, a centralized LLM guardrail service (input validation, output policy checks, PII redaction, schema enforcement), reducing agent policy violations by **70%**.

Client: WEX

03/10/2023 – 30/06/2024

- Built streaming analytics on **Spark Structured Streaming** processing **1.2M+ daily events** (windowing, watermarking, stateful aggregations) for real-time hotel booking metrics.
- Built event-sourced financial reconciliation with idempotent transaction processing for multi-currency merchant accounts across distributed ledger replicas.
- Modernized Kafka → Spark → BigQuery pipeline with CDC and exactly-once delivery, reducing reconciliation lag **65%** across 40+ partner integrations.

Client: Atlassian

14/02/2022 – 30/07/2023

- Designed and led the development of consistent-hashing load balancer initially for insta-switch active-passive elastic search routing, then for active-active Elasticsearch, increasing **Confluence search availability +7% globally**.
- Partnered with deployment engineering teams to architect insta-deploy to all regions from main branch,

cutting **deploy lead time from 1 week to 10 hours.**

Founding Engineer, Packsmith

Jul 2024 – Nov 2024

Packsmith is an AI-driven distribution platform specializing in last mile logistics for e-commerce brands.

- Led end-to-end technical strategy for the distributed reverse logistics platform, partnering with founders and PMs on architecture and milestones.
- Built a priority/vip tagging system for Shopify feeds, improving shipping times for priority orders by 140%.
- Built workflows in Temporal to optimize shipment flow, ensuring accurate stage progression and significantly reducing operational overhead.

Engineering Manager|Staff Engineer, Canva, Sydney, Australia

Nov 2018 - Jan 2022

- Scaled **Canva Search from 20M to 80M MAU** — decomposed a monolithic ingestion pipeline into microservices, added cache deduplication/invalidation, offloaded non-queryable fields from Solr to MySQL, and tuned JVM GC for consistent SLAs; shipped autocorrection, query suggestions, and trending insights end-to-end.
- Built and led a 6-engineer team shipping 7 live-collaboration products in under 2 years, driving **10M+ MAU growth (20% of Canva's total users)** — owned hiring, 1-1s, and reviews while staying hands-on in architecture and code.

Senior Staff Engineer, eBay, San Jose, California

Nov 2014 - Aug 2018

- Headed design of a **dynamic configuration system** and **page assembler** for eBay's core search, letting business teams adapt search UI/UX without engineering involvement — cutting **months off release cycles**.
- Built a **multi-turn dialog SDK** (carousels, media, questions) with **Facebook** and **Google Assistant** integrations, shipping complex conversations in minutes and increasing **Shopbot sales 10x**.
- Created **Regressr**, an open-source regression-testing framework adopted across multiple departments; led project teams of **6+ engineers**, setting technical vision and roadmaps.

Principal Engineer (ICT 5), Apple, Cupertino, California

Sep 2013 - Oct 2014

- Principal engineer for a **globally distributed load balancer** with **99.999% availability** powering **device activations** across the US, Canada, and Europe.
- Oversaw **12 engineers** — project timelines, code reviews, and architectural decisions; built core platform pieces including **config management**, **heartbeat monitoring**, and **deployment orchestration**.

Staff Software Engineer, Walmart, San Bruno, California

Oct 2010 - Sep 2013

- Architect and technical lead for **Walmart Canada's SEO platform** and base-variant management system, reducing **bounce rates by 30%**.
- Built **cloud deployment parallelization** for **300+ applications (80%+ faster deploys)**, an **enterprise-wide config management framework** unifying 200+ microservices, and a **CRUD API auto-generator** saving IaaS teams **8+ months of manual work**.

Earlier roles: Software Engineer at Best Buy (2005–2009), Pearson VUE (2009–2010), Walgreens (2010).

Patents & Publications

5 US patents spanning interactive presentations (US 20230205552, US 20230205808), remote presentation control (US 20230076902), PaaS throughput (US 9342369), and multi-node deployment verification (US 20140109082). Published: Building a Data-Driven Autocorrection System (Canva Engineering Blog), Lessons Learnt from Building Reactive Microservices for Canva Live (Medium/Canva), Introducing Regressr (eBay Tech Blog).